```
x: Integer = 42
y: Float = sin(x)
z: String = "Value"
```

Type Checking

```
assert(x >= 0)
a <- angleCmp(x)

verify(
 0 <= a &&
 a <= 40
)
```

Verification

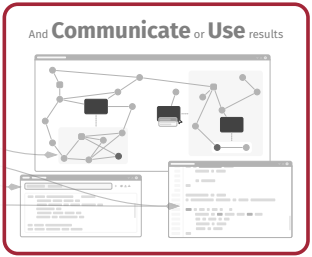
```
for (i in 1:n) {
  sum <- sum + i
}

sum <- n*(n+1)/2
```

Optimization

```
x <- 42
if (x < 0) {
  print("Negative")
}  # Dead Code
```

Linting




```
inp <- read.csv("data.csv")
clean <- filter(inp, a>5)
plot(inp$a, inp$a)
```

Slicing

```
x <- u + 0
print(x)

y <- u + 0
print(y)
```

Refactoring

A Primer on Static Code Analysis

Software Quality Assurance — Static Code Analysis, I | Florian Sihler | December 3, 2025

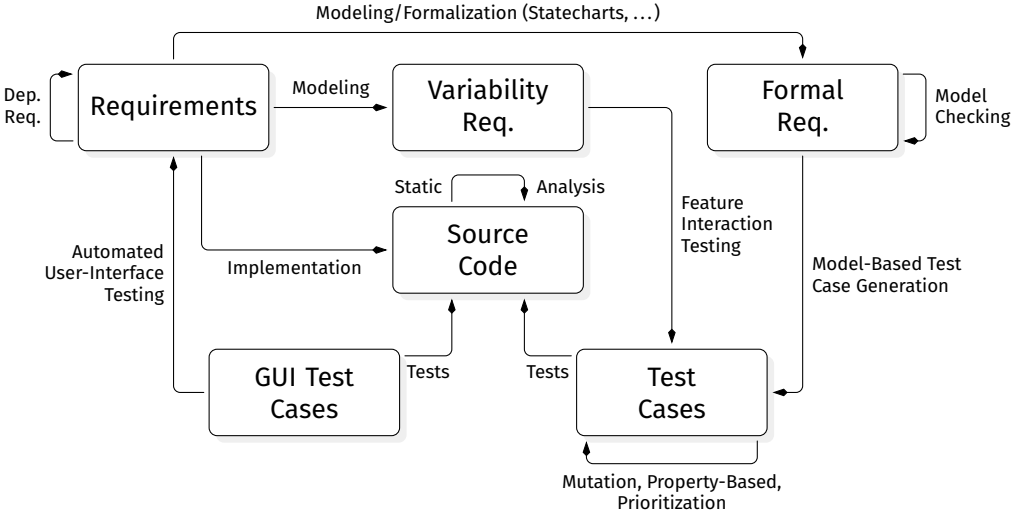
Software Engineering
Programming Languages

universität ulm

1. A first Overview

Embedding a Landscape

Embedding a Landscape



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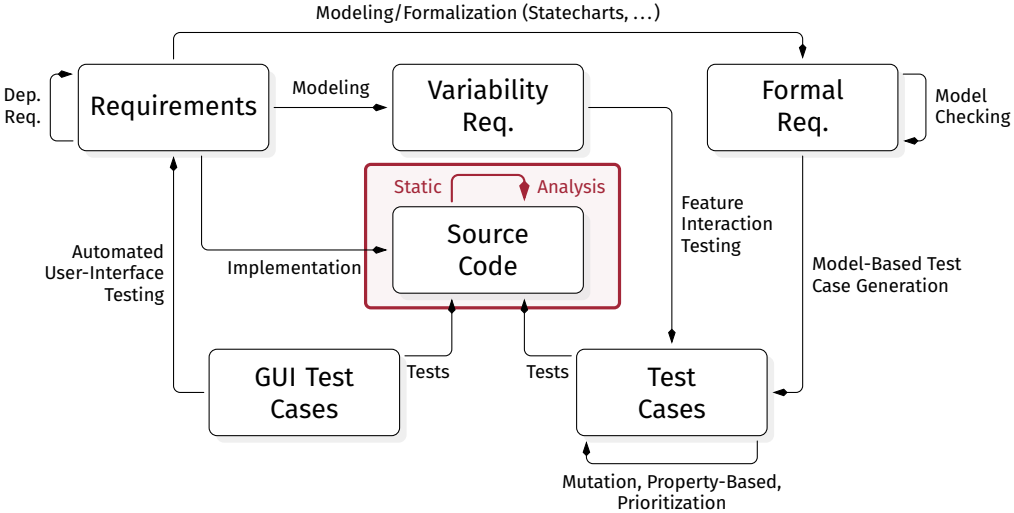

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Discover *syntactic/semantic properties* of programs
without running them.[RY20]

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- Prove correctness
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- Follow coding guidelines / style guides
- Refactoring support
- … (it is fun!)

Let’s find some bugs!

Let’s find some bugs!

Let’s test!

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Dynamic

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Dynamic

Practical

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“Program testing can be used to show the presence of bugs, but never to show their absence!”

Dijkstra[Dij69]

Let's find some bugs!

Edsger W. Dijkstra (1930–2002)
Communications of the ACM

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A First Look

A First Look

They take **Input**

A First Look

And use **Perspectives** (often combined)

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And use **Perspectives** (often combined)

They take **Input**

Textual

A First Look

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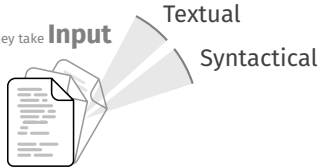

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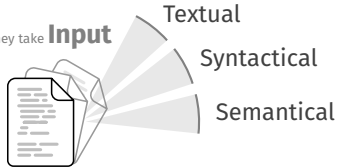

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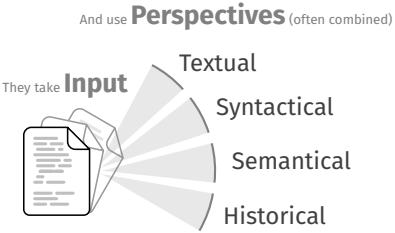
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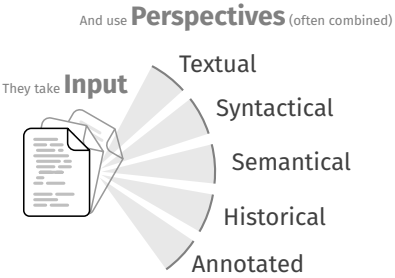
Semantical
Historical

A First Look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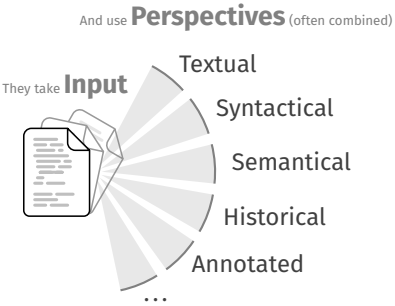
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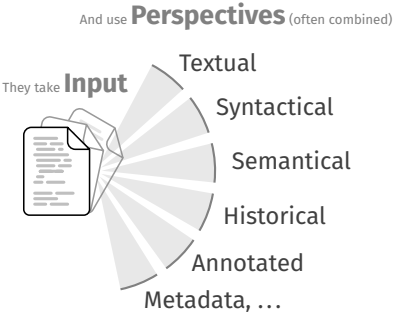
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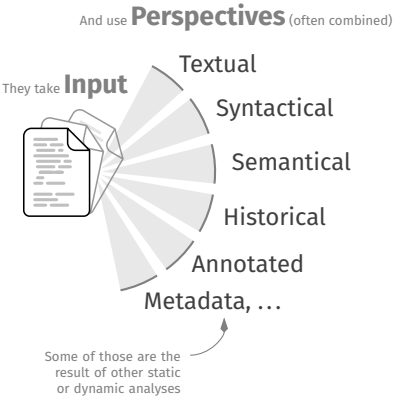
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Some of those are the result of other static or dynamic analyses

A First Look

And use **Perspectives** (often combined)

To apply **Theory**

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To apply **Theory**

Text/Code Search

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To apply **Theory**

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Abstract Domains

A First Look

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They take **Input**

- Textual
- Syntactical
- Semantical
- Historical
- Annotated
- Metadata, …

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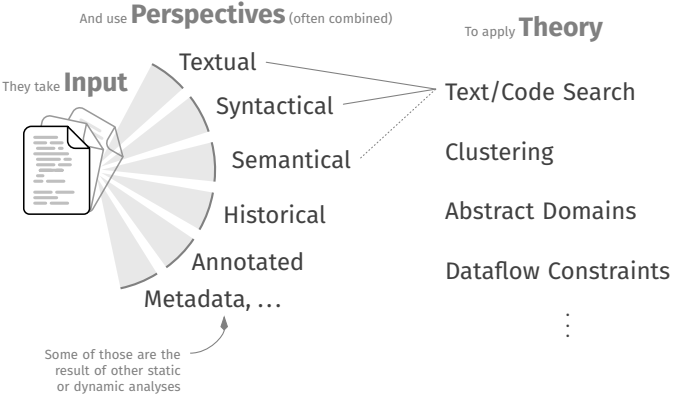
To apply **Theory**

- Text/Code Search
- Clustering
- Abstract Domains
- Dataflow Constraints

⋮

A First Look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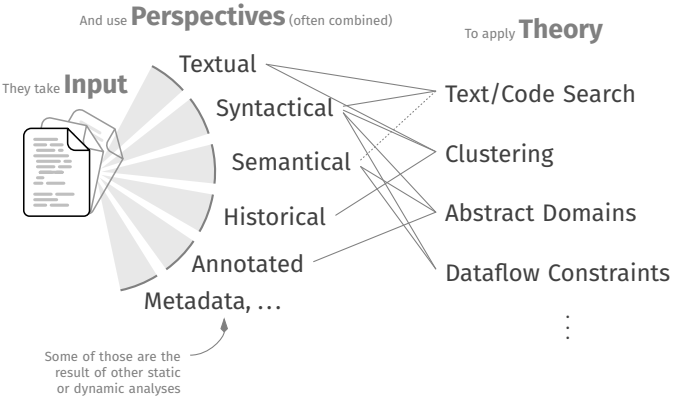
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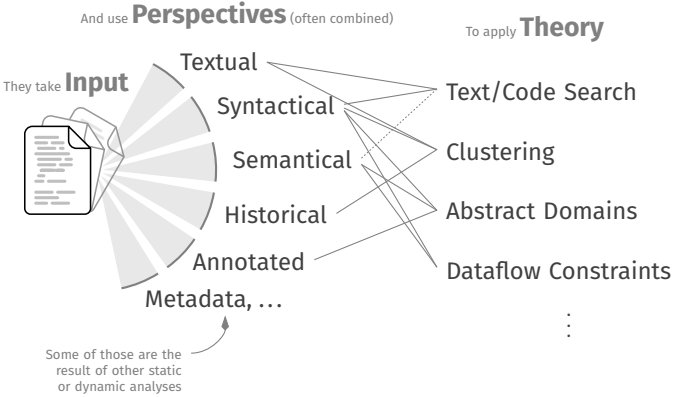

And **Communicate** or **Use** results

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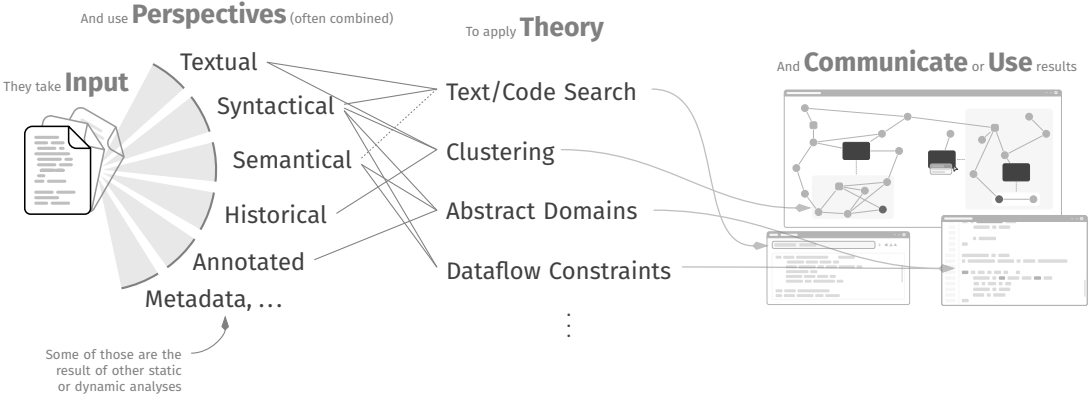
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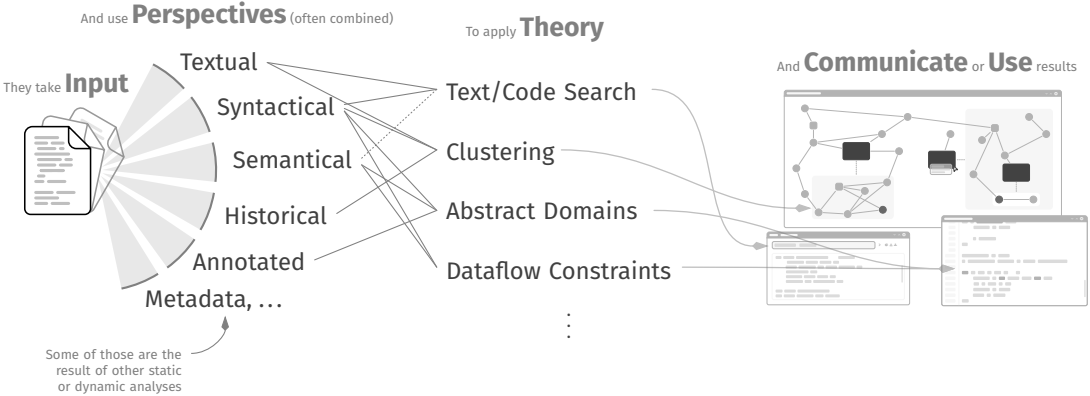

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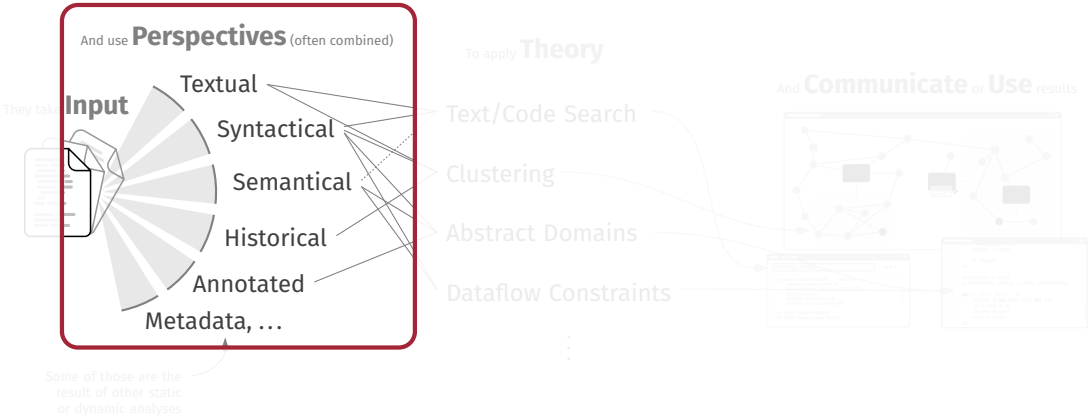
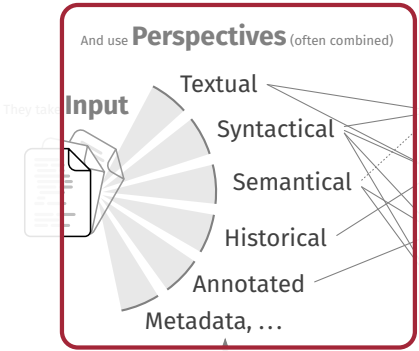


Perspectives

```
x ← sin(y) * z
```

Textual Perspective



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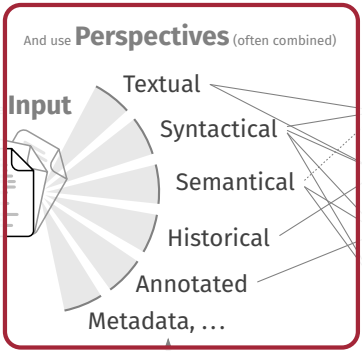


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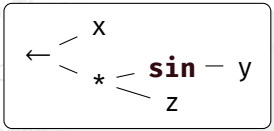
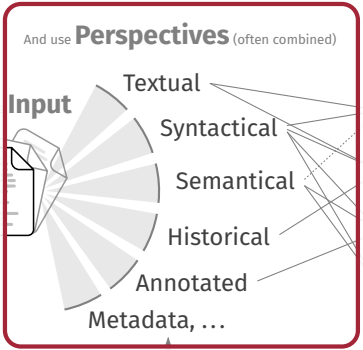


Syntactical Perspective (AST)

Perspectives

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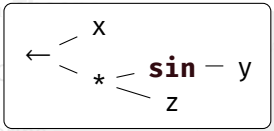

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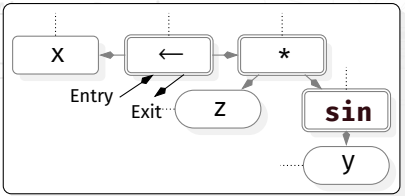
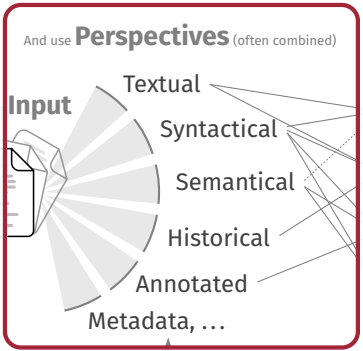


Semantical Perspective (CFG & DFG)

Perspectives

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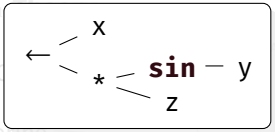
Textual Perspective



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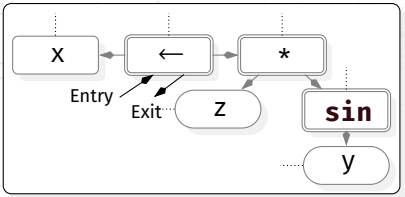

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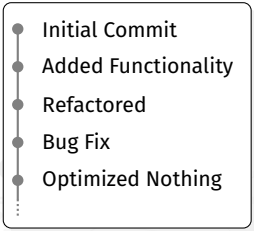
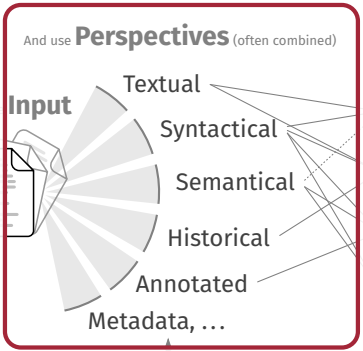


Historical Perspective

Perspectives

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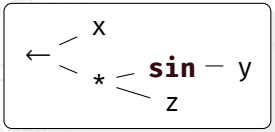
Textual Perspective



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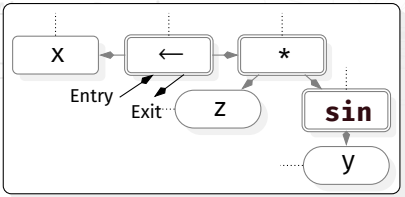

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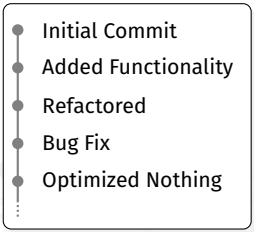
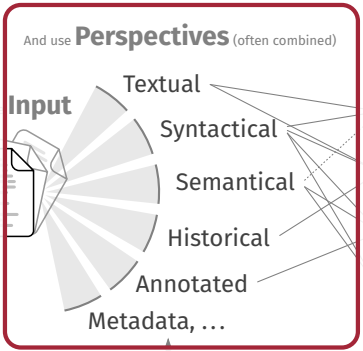


Historical Perspective

```
# Caused CVE 2025-12345
x ← sin(y)
```

Annotated Perspective

Perspectives




```
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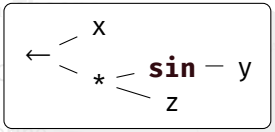
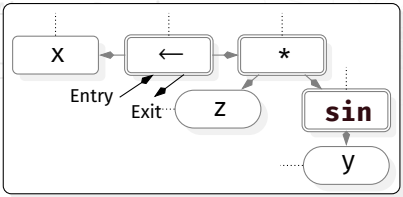
Textual Perspective



Syntactical Perspective (AST)

- Initial Commit
- Added Functionality
- Refactored
- Bug Fix
- Optimized Nothing
- ⋮

Historical Perspective



Semantical Perspective (CFG & DFG)

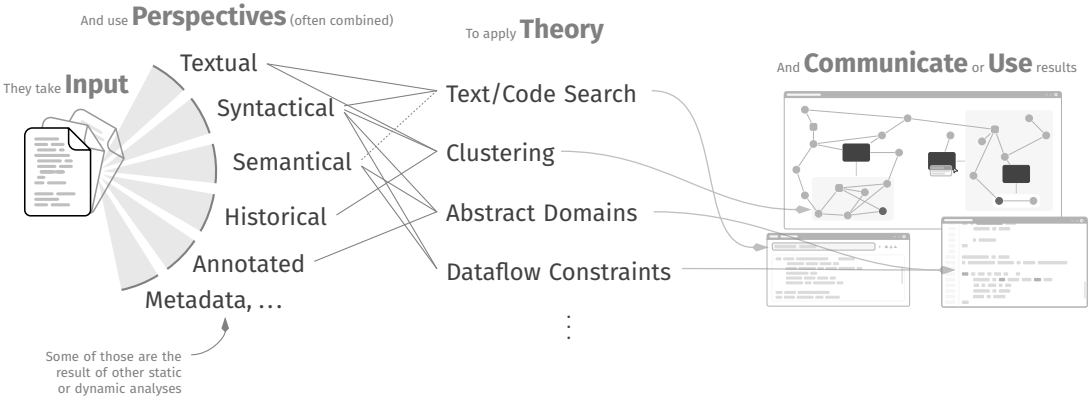
```
# Caused CVE 2025-12345
x ← sin(y)
```

Annotated Perspective

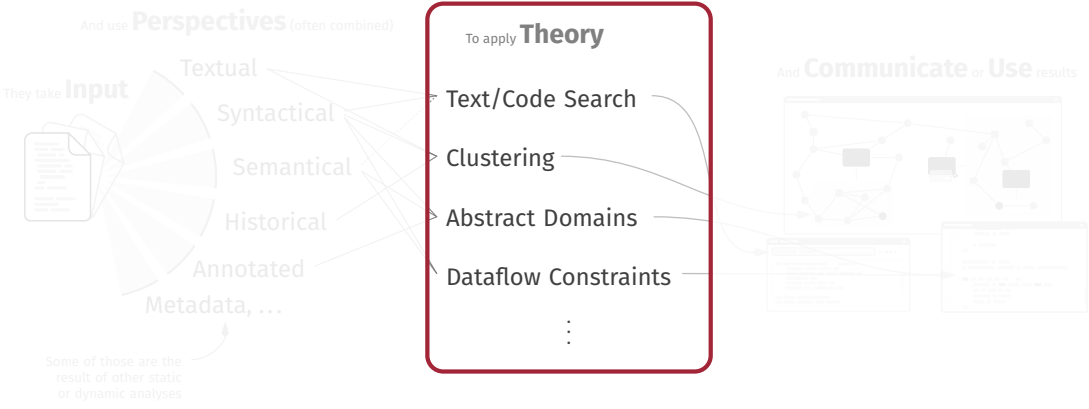
Author: Jane Doe
Code Review: Passed
…

Metadata Perspective

Theory


And use Perspectives (often combined)
To apply Theory
And Communicate or Use results
They take Input
Textual
Syntactical
Semantical
Historical
Annotated
Metadata, …
Text/Code Search
Clustering
Abstract Domains
Dataflow Constraints
⋮
Some of those are the result of other static or dynamic analyses

Theory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And Communicate or Use results

Theory

```
(assignment_expression
  left:  (member_expression
    object:  (call_expression)
))
```

Searching for Patterns

To apply **Theory**

- Text/Code Search
- Clustering
- Abstract Domains
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⋮

Theory

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Searching for Patterns

```
r <- read.csv("data.csv")          load
r <- filter(r, value > 10)         transf.
plot(r$time, r$value)
lines(r$time, r$score)             vis.
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Clustering

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⋮

$$\top = [-\infty \,..\, \infty]$$

$$\bot = \varnothing$$

$$\bigsqcup_k [\,\ell_k \,..\, h_k\,] = [\min(\ell_k) \,..\, \max(h_k)]$$

$$\bigsqcap_k [\,\ell_k \,..\, h_k\,] = [\max(\ell_k) \,..\, \min(h_k)]$$

Abstract Domains

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Clustering

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⋮

```
x ← 2 * rnd(0,1)      x ∈ [0,2]
y ← x + 1             y ∈ [1,3]
z ← y + abs(u)        z ∈ [1,∞]
```

Abstract Domains

Theory

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$\mathtt{z} \leftarrow \mathtt{y} + \mathbf{abs}(\mathtt{u})$ $\quad z \in [1,\infty]$

$\prod_k [\ell_k \,..\, h_k] = [\max(\ell_k) \,..\, \min(h_k)]$

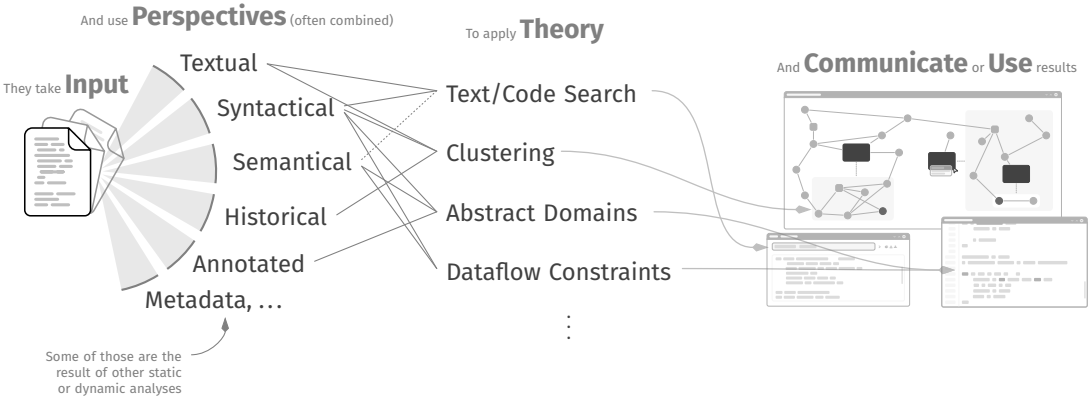
Abstract Domains

$$\mathbf{In}_n = (\mathbf{Out}_n - \mathbf{Kill}_n) \cup \mathbf{Gen}_n$$

$$\mathbf{Out}_n = \begin{cases} \mathbf{BI} & n \text{ is End} \\ \bigcup_{m \in \mathrm{succ}(n)} \mathbf{In}_m & \text{else} \end{cases}$$

Dataflow Constraints

Application



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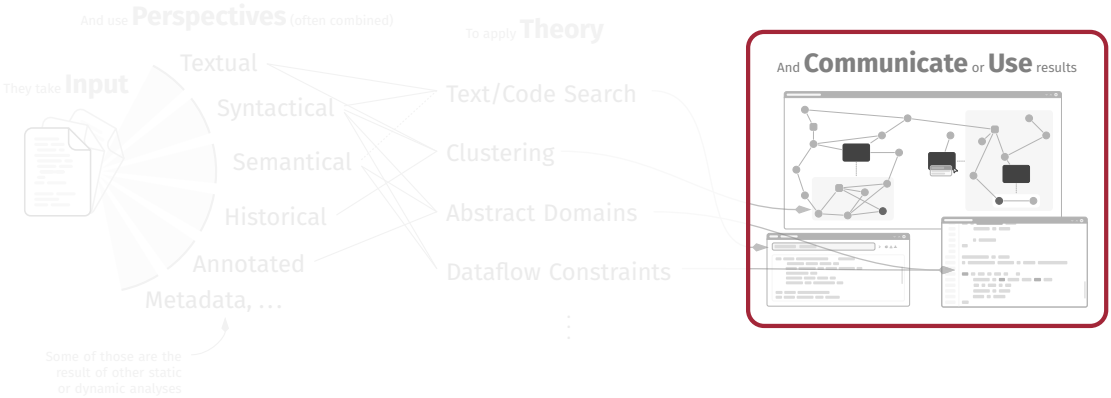

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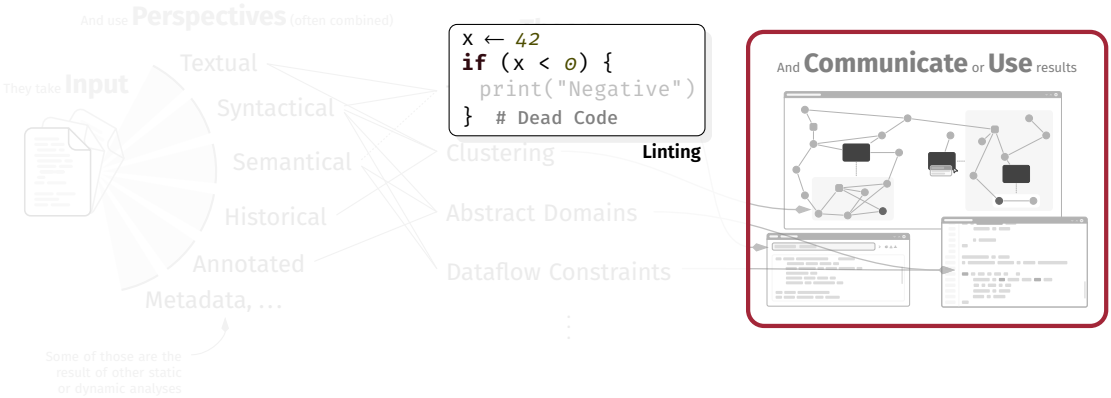

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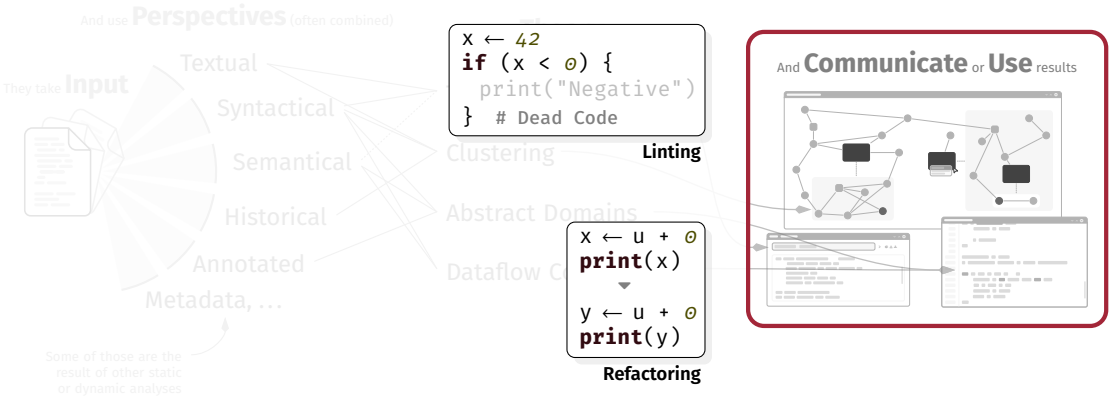


Application

```
x: Integer = 42
y: Float = sin(x)
z: String = "Value"
```

Type Checking


```
x ← 42
if (x < 0) {
  print("Negative")
}  # Dead Code
```

Linting

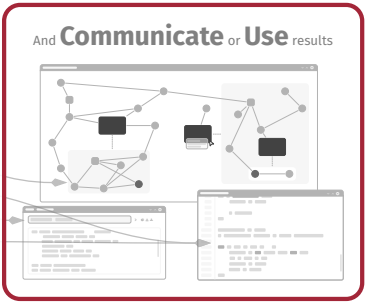
Clustering

Abstract Domains

Dataflow C

```
x ← u + 0
print(x)
▼
y ← u + 0
print(y)
```

Refactoring



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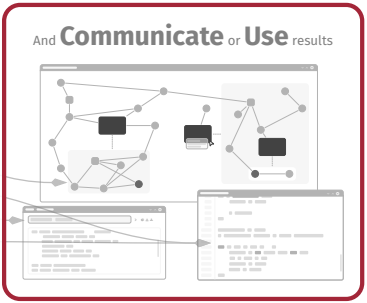
```
inp ← read.csv("data.csv")
clean <- filter(inp, a>5)
plot(inp$a, inp$a)
```

Slicing



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Type Checking

```
for (i in 1:n) {
  sum ← sum + i
}
sum ← n*(n+1)/2
```

Optimization

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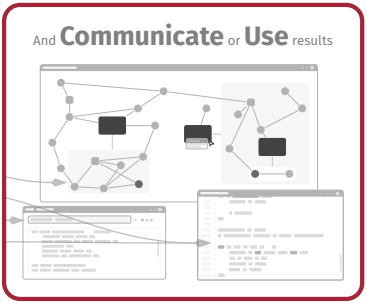

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```

Type Checking

```
assert(x >= 0)
a ← angleCmp(x)

verify(
 0 <= a &&
 a <= 40
)
```

Verification

```
for (i in 1:n) {
  sum ← sum + i
}
▼
sum ← n*(n+1)/2
```

Optimization

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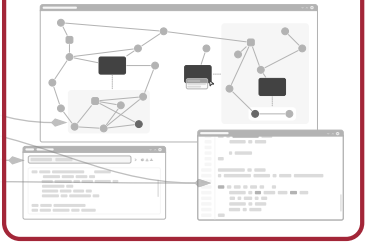
Refactoring

And **Communicate** or **Use** results



2. Linting Origins

A User-Driven History: Linting

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Stephen C. Johnson (1944)
Faces of OS

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Which linters do you know?
⟹ Part 3

A Word on “Linting”

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In the following, we explore some common linting tasks.

Style Checks

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  - …

Style guides can be, language, organization, domain, or even project specific!

Good Code, Bad Code

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```java
int timeToReadBook = (220 * 10) / 60;
```

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Apples? Oranges?

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Apples? Oranges?

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final int PAGES = 220;
final int MINUTES_PER_PAGE = 10;
final int MINUTES_PER_HOUR = 60;

int timeToReadBook = (PAGES * MINUTES_PER_PAGE) /
                     MINUTES_PER_HOUR;
```

And Besides Synthetical Examples?

http://support.apple.com/kb/HT6147

And Besides Synthetical Examples?

C

libsecurity_ssl/lib/sslKeyExchange.c

```c
static OSStatus
SSLVerifySignedServerKeyExchange(SSLContext *ctx, bool isRsa, SSLBuffer signedParams,
                                 uint8_t *signature, UInt16 signatureLen)
{
// ...
    if ((err = SSLFreeBuffer(&hashCtx)) != 0)
        goto fail;

    if ((err = ReadyHash(&SSLHashSHA1, &hashCtx)) != 0)
        goto fail;
    if ((err = SSLHashSHA1.update(&hashCtx, &clientRandom)) != 0)
        goto fail;
    if ((err = SSLHashSHA1.update(&hashCtx, &serverRandom)) != 0)
        goto fail;
    if ((err = SSLHashSHA1.update(&hashCtx, &signedParams)) != 0)
        goto fail;
        goto fail;
    if ((err = SSLHashSHA1.final(&hashCtx, &hashOut)) != 0)
        goto fail;
// ...
fail:
    SSLFreeBuffer(&signedHashes);
    SSLFreeBuffer(&hashCtx);
    return err;

}
```

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Wellp!

Now, *err* can be 0, signaling a valid signature!

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```yaml
rules:
  - id: add-if-braces
    pattern: if ($COND) $BODY
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    rewrite: if ($COND) { $BODY }
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- Examples:

Style Checks: Summary

- How to do style checks?
  - Mostly through Regular Expressions or AST pattern matching
  - Many also offer fixes/suggestions
  - For example (with ast-grep, slightly simplified):

yaml

```yaml
rules:
  - id: add-if-braces
    pattern: if ($COND) $BODY
    constraints: $BODY: not:
          kind: block
    rewrite: if ($COND) { $BODY }
```

- Usually, they are heavily configurable to fit project needs
- Likewise, they are usually only syntactic, without semantic info
  (although many offer more features)
- Examples: checkstyle, pylint, ESLint Sylistic, ...

Please consider using `.git-blame-ignore-revs`.

Unconventional Patterns and Bugs

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Unconventional Patterns and Bugs

- Various Categories:

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Error Mining

See also:

[HP04] David Hovemeyer and William Pugh. Finding bugs is easy (Association for Computing Machinery, 2004)
[Bre12] Alexander Breckel. Error mining: Bug detection through comparison with large code databases (MSR, 2012)

Error Mining

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for(j = 0; j < -n; j++) array[i] = j;

for(i = 0; i <  k; i++)   arr[i] = i;
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- We try to find code that has been *almost* written like this!
- If many do something similar, but one does it differently, it might be a bug
- Usually relies on AST patterns with minimal binding information

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3. More Sophisticated Fronts

A Best Friend for Life: Control Flow Graph (CFG) [CT22]

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Directed graph representing all possible execution paths of a program. Usually uses *basic blocks* as nodes.

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int a = 42;
while(a > 0) {
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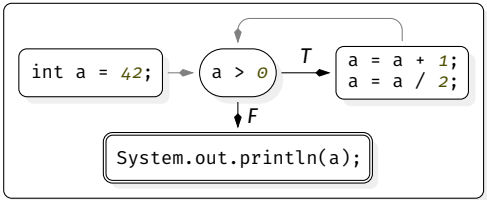

CFG

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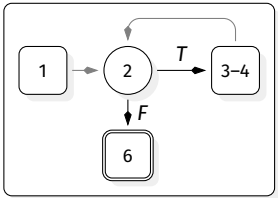

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[CT22] K.D. Cooper and L. Torczon. Engineering a Compiler (Elsevier Science, 2022)

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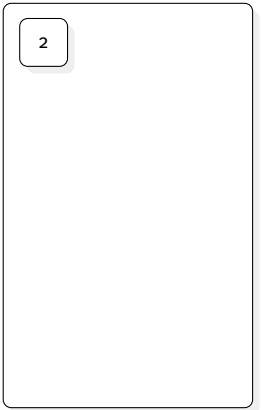

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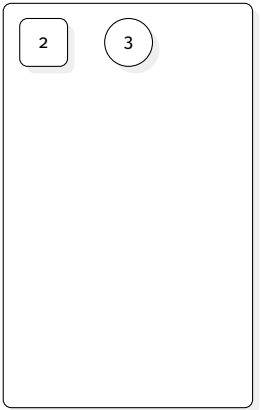

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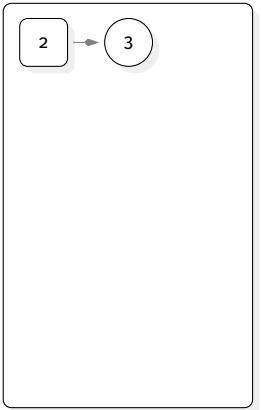

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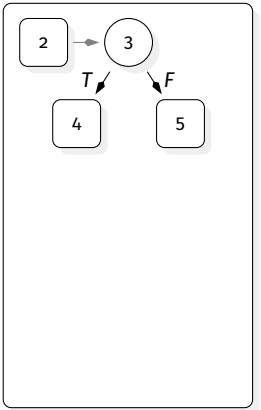

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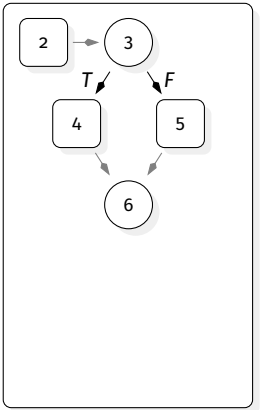

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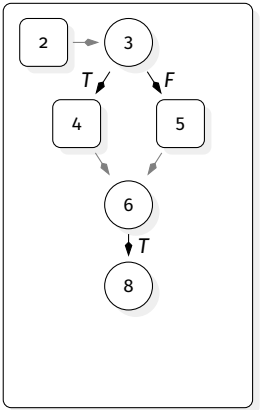

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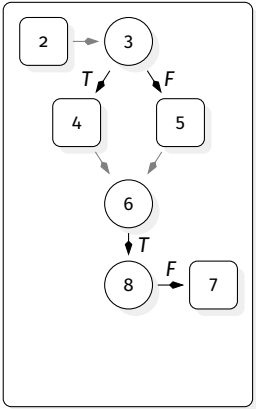

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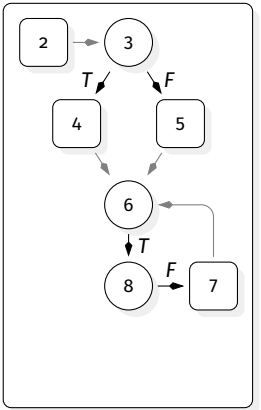

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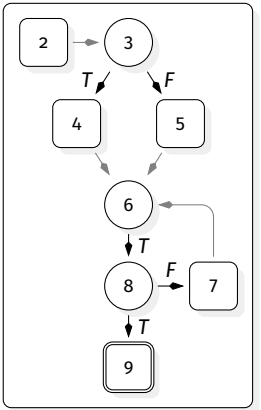

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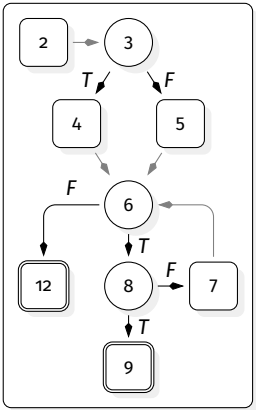

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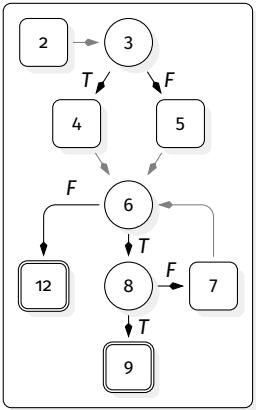

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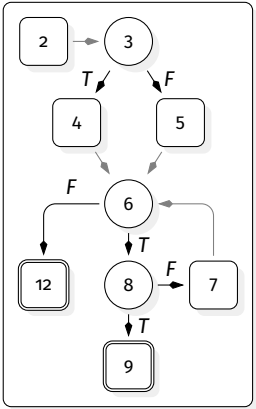

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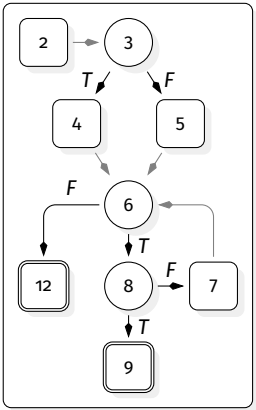

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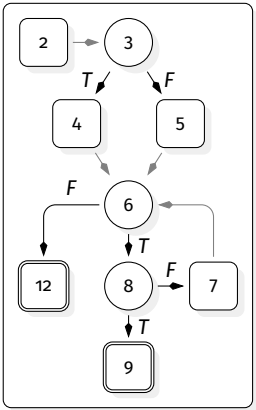
- How to find infinite loops?
  E.g., nodes that cannot reach an exit node.



CFG

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x = 42
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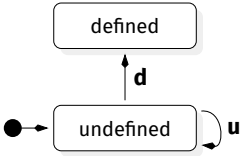

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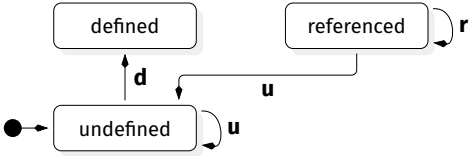

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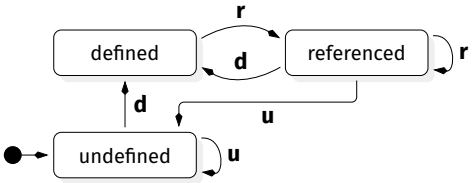

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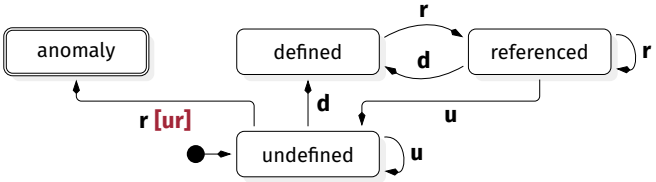

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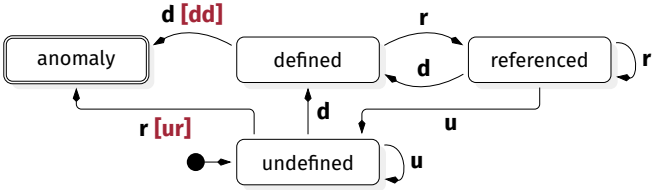

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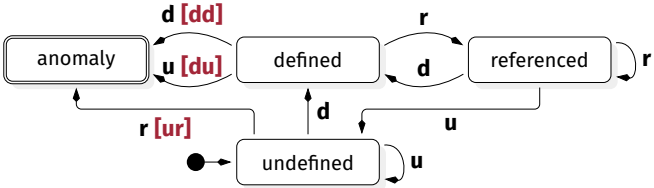

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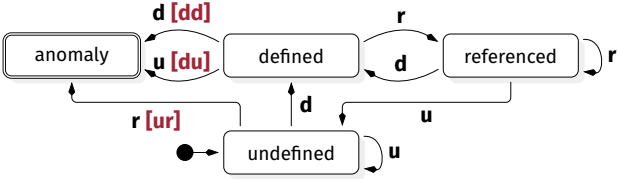

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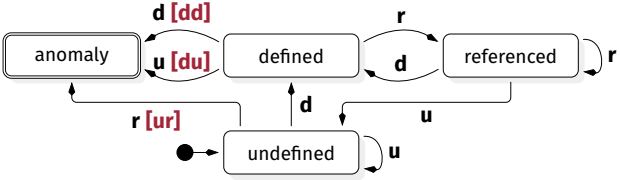

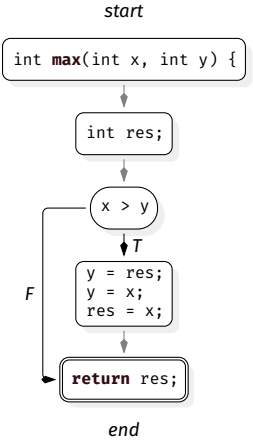

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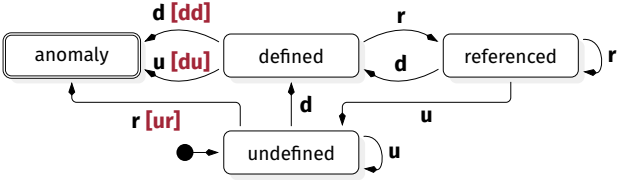
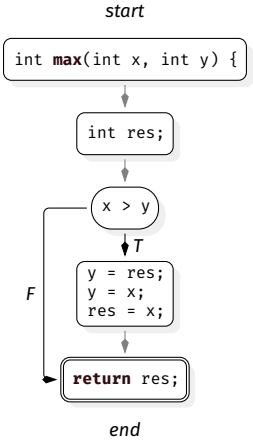


x y res



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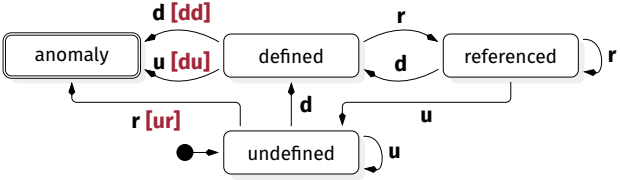


x	y	res
u	u	

start

```
int max(int x, int y) {
```

```
int res;
```

```
x > y
```

T

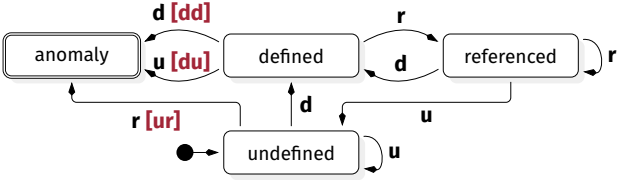
```
y = res;
y = x;
res = x;
```

F

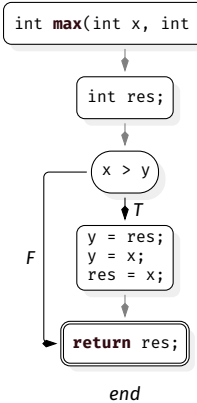
```
return res;
```

end

Let's Run an Example!



	x	y	res
start	u	u	
`int max(int x, int y) {`	d	d	



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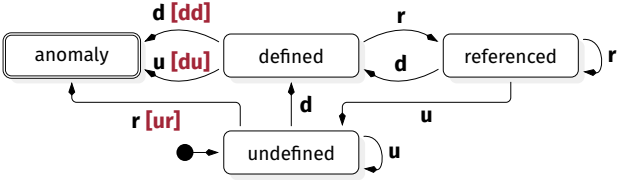

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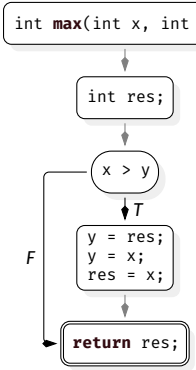

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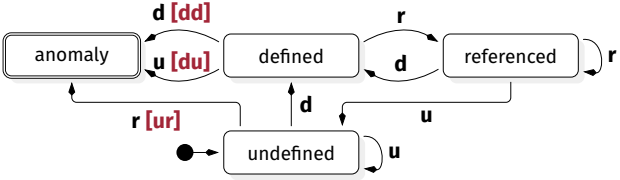

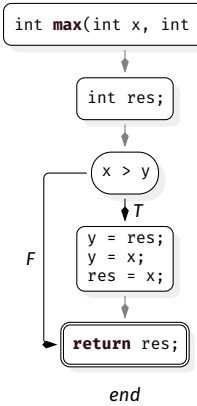

x	y	res
u	u	
d	d	
		u
r	r	

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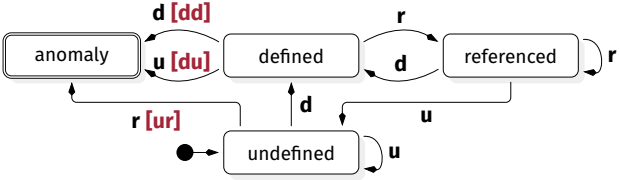

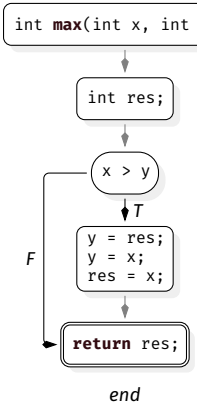

x	y	res
u	u	
d	d	
		u
r	r	
	d	r

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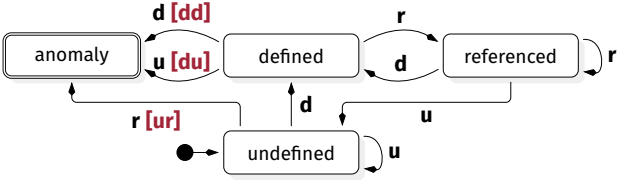

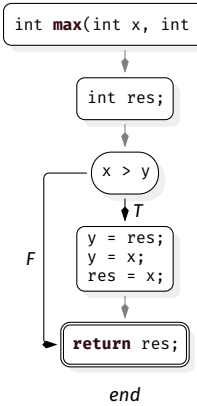

x	y	res
u	u	
d	d	
		u
r	r	
	d	r
r	d	

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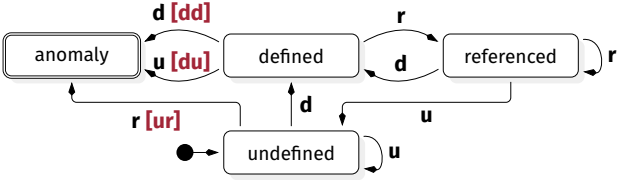

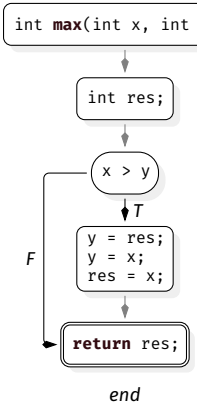

x	y	res
u	u	
d	d	
		u
r	r	
	d	r
r	d	
r		d

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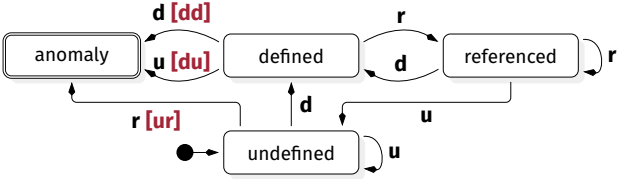

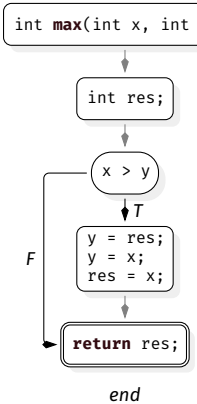

x	y	res
u	u	
d	d	
		u
r	r	
	d	r
r	d	
r		d
		r

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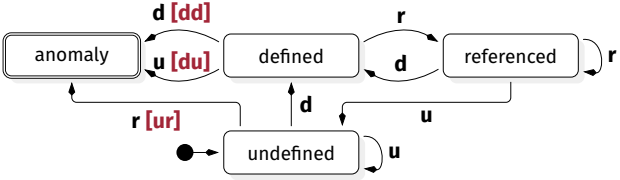

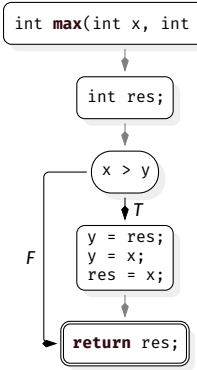

	x	y	res
start	u	u	
`int max(int x, int y) {`	d	d	
`int res;`			u
`x > y`	r	r	
`y = res;`		d	r
`y = x;`	r	d	
`res = x;`	r		d
`return res;`			r
end	u	u	u

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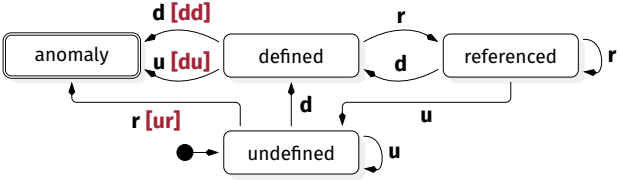

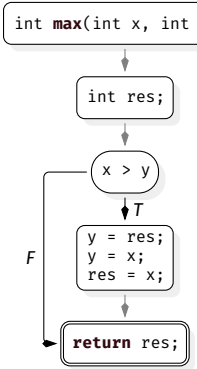

	x	y	res
start	u	u	
int max(int x, int y) {	d	d	
int res;			u
x > y	r	r	
True			
y = res;		d	r
y = x;	r	d	
res = x;	r		d
return res;			r
end	u	u	u

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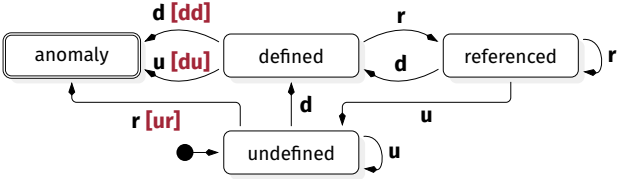

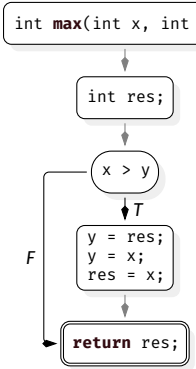

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x > y	r	r	
True			
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y = x;	r	d	
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return res;			r
end	u	u	u

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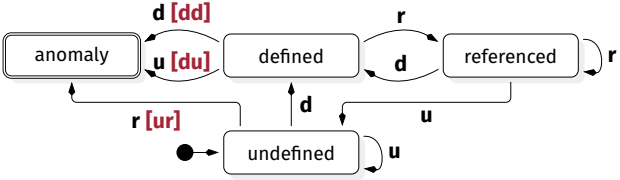
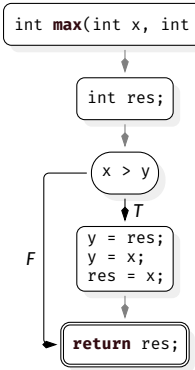
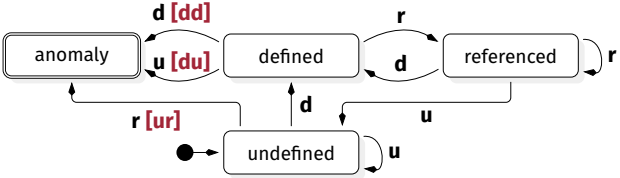


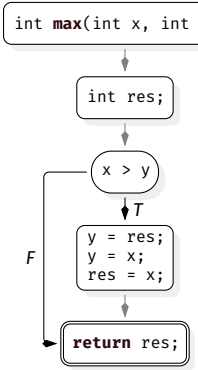


x	y	res
u	u	
d	d	
		u
r	r	
r	d	r
r	d	d
		r
u	u	u

- **[dd]**: Unnecessary definition

Let's Run an Example!





	x	y	res
start	u	u	
int max(int x, int y) {	d	d	
int res;			u
x > y	r	r	
True		d	r
	r	d	
	r		d
			r
end	u	u	u

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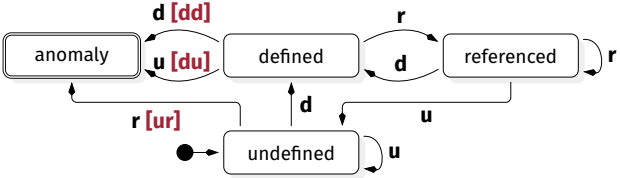

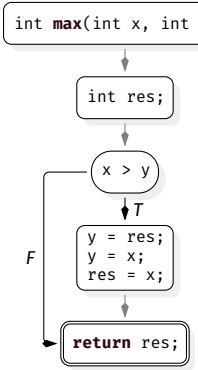

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int max(int x, int y) {	d	d	
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x > y	r	r	
T: y = res;		d	r
y = x;	r	d	
res = x;	r		d
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True

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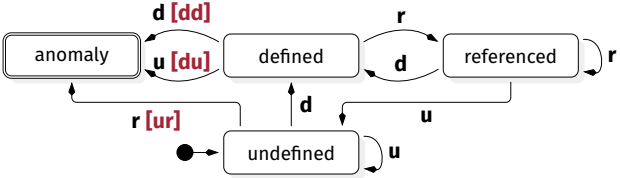

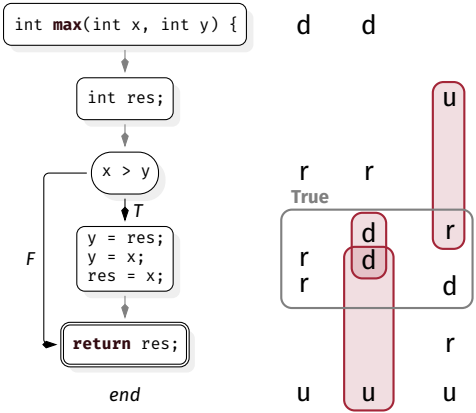

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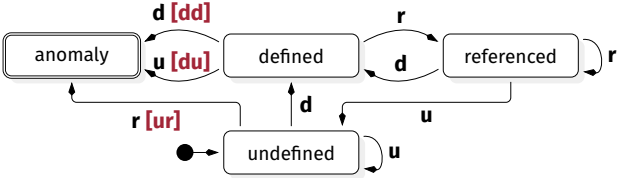

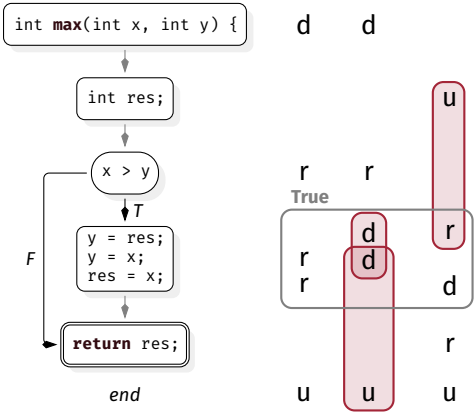

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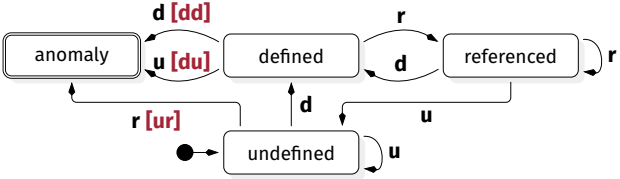

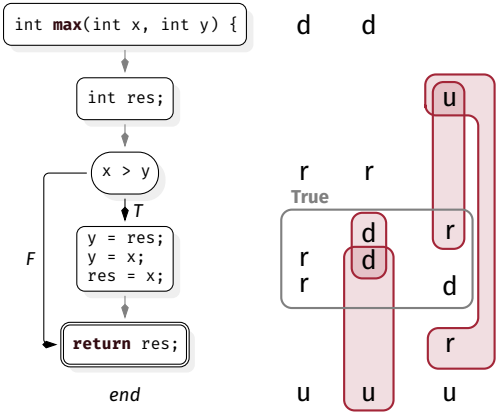

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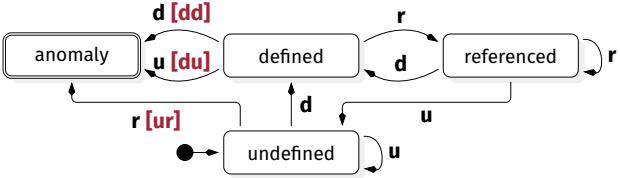

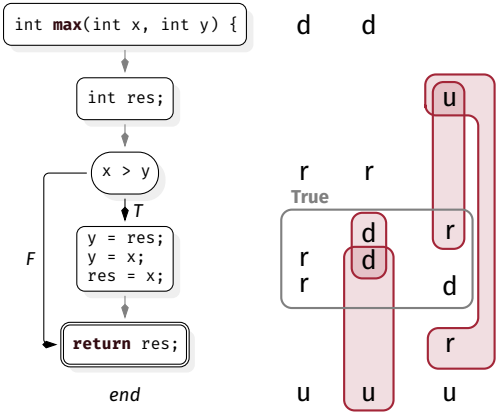

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- Consider all paths through the CFG!

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- A plethora of techniques try to address these challenges from various perspectives
- One prominent technique is *abstract interpretation*[Cou21]
  We'll tackle this in the second lecture!

4. Important Terminology

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- **Precision:** $\mathbf{TP}/(\mathbf{TP} + \mathrm{FP})$ ("how many false alarms")
- **Recall:** $\mathbf{TP}/(\mathbf{TP} + \mathrm{FN})$ ("how many errors did we find")

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    You better want completeness.

5. Outlook

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We will address these questions in the next lecture(s)!

References I

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